Software Requirements Document (SRD)

Project Name

Automated HVAC Duct Detection and Annotation Web Application

1. Introduction

1.1 Purpose

The purpose of this web application is to allow users to upload HVAC mechanical drawing PDFs, automatically detect ductwork, extract duct dimensions and lengths, and generate annotated output drawings through a browser-based interface.

The system is intended to reduce manual review effort and provide an accessible, visual, and interactive platform for HVAC drawing analysis.

1.2 Scope

The web application will:

- allow users to upload HVAC mechanical drawing PDFs
- process uploaded drawings on the server
- detect supply and return ducts from the plan
- extract duct size labels from nearby annotations
- estimate duct lengths using drawing scale and geometry
- generate annotated output drawings
- allow users to preview, review, and download results
- optionally export structured data in JSON or CSV format

The product is a **web-based document analysis platform** focused on duct detection and markup generation.

1.3 Intended Users

- HVAC engineers
- MEP consultants
- estimators
- permit drawing reviewers
- QA/documentation teams
- project coordinators reviewing mechanical plans

1.4 Reference Context

The provided sample indicates:

- input: mechanical floor plan PDF
- output: annotated visual image highlighting duct runs
- the expected output is primarily a visual markup artifact suitable for validation.

2. Product Vision

The system will provide a browser-based workflow where a user can:

1. upload a mechanical drawing PDF
2. trigger automated duct detection
3. view the annotated result directly in the browser
4. inspect detected ducts and labels
5. download the final annotated PNG/PDF and structured data exports

The product should feel like a lightweight engineering review tool rather than a generic file uploader.

3. Business Objective

The business objective is to provide a scalable and user-friendly web application for automated HVAC duct markup so engineering and estimating teams can:

- review plans faster
- reduce manual duct tracing
- standardize markup output
- improve turnaround time
- centralize drawing analysis in an accessible browser interface

4. Problem Statement

HVAC mechanical drawings are visually dense and require manual review to identify duct runs, dimensions, and lengths. Traditional manual markup is slow, inconsistent, and difficult to scale across multiple projects.

A web application is needed that can:

- ingest PDF drawings through a browser
- process them automatically on the backend
- visualize the detected duct network
- present measurement and annotation results interactively
- support downloading output deliverables

5. Product Overview

The system is a **web application** consisting of:

- a **frontend UI** for file upload, job status, preview, and download
- a **backend processing service** for PDF parsing, duct detection, measurement, and annotation
- optional **persistent storage** for uploaded files and generated outputs
- optional **database storage** for job metadata and results

The final deliverables are:

- annotated visual output
- downloadable output files
- optional structured duct data

6. User Roles

6.1 Standard User

A user who uploads a drawing, triggers processing, views results, and downloads outputs.

6.2 Admin User

A user who can monitor processing jobs, review failures, inspect logs, and manage system configuration.

7. Functional Requirements

7.1 Authentication and Access

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The web app may support user authentication.

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Authenticated users shall have access to their own uploaded files, jobs, and outputs.

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Admin users shall be able to view system-wide processing logs and job statuses.

7.2 File Upload and Input Management

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The system shall provide a web interface for uploading PDF files.

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The system shall validate uploaded files to ensure they are supported PDF documents.

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The system shall allow the user to upload:

- a single PDF
- multi-page PDFs

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The system shall store the uploaded file temporarily or persistently based on configuration.

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The system shall display metadata for uploaded files, such as:

- file name
- upload time
- number of pages, where available
- processing status

7.3 Job Creation and Processing

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The system shall create a processing job after a file is uploaded.

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The system shall process jobs asynchronously so the user interface remains responsive.

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The system shall display job states such as:

- uploaded
- queued
- processing
- completed
- failed

FR-12

The system shall allow the user to initiate processing from the web interface.

FR-13

The system shall support processing of:

- all pages
- a selected page
- selected page ranges

7.4 Drawing Parsing and Plan Analysis

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The system shall identify the main drawing region and distinguish it from:

- title block
- notes section
- legends
- margins
- empty space

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The system shall support PDFs containing:

- machine-readable vector content
- image-based or scanned content, where feasible

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The system shall extract drawing text and annotation text with coordinate information.

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The system shall maintain coordinate mapping between:

- PDF coordinates
- rendered image coordinates
- annotation overlay coordinates

7.5 Duct Detection

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The system shall detect ductwork in the HVAC mechanical drawing.

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The system shall identify major duct runs and segments.

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The system shall support detection of:

- supply ducts
- return ducts

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The system shall identify duct segments between geometric change points such as:

- elbows
- branches
- transitions
- junctions

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The system shall support at minimum:

- rectangular ducts
- circular ducts where notation supports interpretation

7.6 Label and Dimension Extraction

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The system shall extract duct size labels from text annotations near detected duct segments.

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The system shall recognize common dimension formats such as:

- `24x12`
- `24 x 12`
- `Ø12`
- `12 DIA`

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The system shall associate extracted labels with the most likely duct segment.

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The system should compute a confidence score for each text-to-duct association.

7.7 Measurement

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The system shall detect explicit drawing scale where available. The sample sheet includes a printed drawing scale.

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The system shall estimate duct segment lengths using:

- detected geometry
- scale conversion
- or explicit annotation where available

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The system shall present length values in a configurable unit format.

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The system shall document the measurement convention used, such as centerline length.

7.8 Result Visualization

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The web app shall provide an in-browser result preview.

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The preview shall display the original drawing with highlighted duct overlays.

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The preview shall display labels for detected ducts, including:

- duct size
- segment ID
- length, where available

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The web app should allow zoom and pan on the annotated drawing.

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The web app should provide a side panel or summary view listing detected duct segments and their properties.

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The web app should distinguish low-confidence detections visually.

7.9 Download and Export

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The system shall allow the user to download the annotated result as PNG.

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The system shall allow the user to download the annotated result as PDF.

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The system should allow structured export as:

- JSON
- CSV

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The system shall preserve output files for a configurable retention period.

7.10 Debug and Review Support

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The system should provide a debug mode for admins or advanced users.

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Debug mode may show:

- detected plan crop
- detected text boxes
- candidate duct geometries
- label association regions
- rejected detections

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The system shall log processing results and errors for each job.

7.11 Job History

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The system shall provide a job history view for each user.

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The user shall be able to reopen a completed job and review/download prior results.

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The admin shall be able to inspect failed jobs and related logs.

8. User Journey

8.1 Primary Flow

1. User opens the web app
2. User uploads a PDF drawing
3. User selects page or processing options
4. User starts processing
5. Web app shows job status
6. Backend processes drawing
7. User views annotated result in browser
8. User downloads annotated PNG/PDF and optional JSON/CSV

8.2 Failure Flow

1. User uploads invalid or unreadable PDF
2. System rejects file or marks job as failed
3. User receives a readable error message
4. Admin can inspect logs if needed

9. Non-Functional Requirements

9.1 Performance

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The web app should upload and register a file within a few seconds under normal network conditions.

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The backend should process a standard single-sheet HVAC drawing within an acceptable target time, such as under 1 minute on typical infrastructure.

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The UI shall remain responsive while processing occurs asynchronously.

9.2 Scalability

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The system should support multiple concurrent users and multiple queued jobs.

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The job-processing backend should be horizontally scalable.

9.3 Usability

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The interface shall be simple and intuitive for non-developer users.

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The annotated preview shall be readable and navigable.

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The file upload and result download workflow shall require minimal training.

9.4 Reliability

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The system shall handle failed jobs gracefully without crashing the application.

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The system shall preserve job status and error details for troubleshooting.

9.5 Security

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Uploaded files shall be processed securely.

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Users shall only be able to access files and results they are authorized to view.

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The system shall use HTTPS in production.

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Sensitive project files should be retained only according to configured policy.

9.6 Maintainability

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The application shall follow a modular design separating:

- frontend
- API/backend
- processing engine
- file storage
- job management

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The system should support future expansion to other MEP object types.

9.7 Portability and Deployment

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The system should be deployable on cloud or on-premise infrastructure.

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The application should be containerized using Docker.

10. System Components

The web app shall include the following major components:

- **Frontend UI**
    - file upload
    - job status
    - preview
    - results panel
    - downloads

- **Backend API**
  - upload endpoint
  - job management
  - result retrieval
  - authentication integration
- **Processing Engine**
  - PDF parsing
  - duct detection
  - text extraction
  - measurement
  - annotation rendering
- **Storage**
  - file/object storage for PDFs and outputs
  - database for job metadata and results
- **Queue/Worker Layer**
  - asynchronous job execution

11. External Interfaces

11.1 Frontend

Browser-based web interface.

11.2 Backend API

REST API for:

- upload
- create job
- get job status
- get preview/result
- download outputs

11.3 File Interfaces

Input:

- PDF

Output:

- PNG
- PDF
- JSON
- CSV
- logs

12. Assumptions

- Users will access the system via a modern web browser
- Drawings will be uploaded as PDFs
- Duct labels are present near duct runs in most usable drawings
- The desired output is a visually annotated drawing, based on the provided sample.
- Processing can occur asynchronously rather than instantly in the request cycle

13. Constraints

- PDF drawing formats may vary significantly by project
- Scanned plans may have lower text extraction quality
- Dense linework may affect geometry detection
- Some duct segments may lack clearly readable size labels
- Browser preview performance may degrade for very large drawings if not optimized

14. Acceptance Criteria

The web application will be accepted if it can:

1. allow a user to upload the sample HVAC PDF through the browser
2. process the file successfully
3. detect and highlight the major duct runs
4. display an annotated result in the browser
5. allow download of annotated PNG/PDF
6. extract and show meaningful duct dimension and length information for a significant portion of detected segments
7. handle failed uploads or failed processing gracefully
8. maintain job status visibility throughout the process

15. Future Enhancements

- manual correction tools in browser
- interactive segment editing
- measurement override UI
- project-level workspace management
- team collaboration/comments
- OCR enhancement for scanned drawings
- BOQ/takeoff generation
- BIM/CAD integration

16. Summary

This SRD defines a **web application** for automated HVAC duct detection and annotation. The application enables users to upload HVAC mechanical drawing PDFs, process them asynchronously on the backend, preview annotated results in the browser, and download visual and structured outputs. The system is centered around a practical engineering-review workflow and aligns with the sample PDF input and marked-up output image you shared.